

Itzel Salazar

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EDUCATION

The University of Texas at El Paso

Master of Science in Software Engineering (Secure Cyber Systems)

May 2025

The University of Texas at El Paso

Bachelor of Business Administration in Operations and Supply Chain Management

Fall 2019

PROJECTS

Fashion Data Vault | *Python, FastAPI, Next.js, TypeScript, Tailwind, PostgreSQL, spaCy, Git*

- Built a trend intelligence pipeline that ingests and normalizes RSS articles from Vogue, WWD, Business of Fashion, Elle, and Fashionista, with deduplication to prevent duplicate article storage.
- Designed trend aggregation logic using daily snapshots to accurately track trend frequency over time.
- Developed a custom spaCy NLP pipeline using EntityRuler and Matcher to extract seasons, brands, materials, and trend patterns from RSS articles.
- Built domain specific vocabularies and pattern rules for trend detection by manually analyzing over 300+ fashion articles and identifying common descriptors, materials, and terminology.
- Deployed a full-stack Next.js and FastAPI application on Vercel and Railway with a custom domain.

Rat Brain Atlas Visualization Tool | *Python, MongoDB, Next.js, Typescript, Tailwind, Flask, Git*

- Developed the homepage and navigation for an interactive rat brain atlas which allows users to explore neuroanatomical regions via a coordinate-based interface.
- Developed region specific collaborative note taking functionality for users to create and save annotations associated with specific brain regions for research use.
- Assisted in the development of layered SVG brain renderings, enabling toggleable overlays such as coordinate mappings for better anatomical reference.
- Contributed to the design of authentication and role-based access controls.
- Worked closely with UTEP Neuroscience faculty to convert their research requirements into a workable digital tool for neuroanatomical mapping and studies.

TECHNICAL SKILLS

Languages: Python, Java, Typescript, SQL, JavaScript

Frameworks: FastAPI, Next.js, Flask, Tailwind CSS

Developer Tools: Git, Docker, Vercel, Railway

Databases: PostgreSQL, MongoDB

NLP/ML: spaCy (EntityRuler, Matcher)

Libraries: pandas, psycpg3

VOLUNTEER EXPERIENCE

Microsoft TEALS Program | *August 2023 - May 2024*

- Co-taught the Code.org AP Computer Science curriculum through the Microsoft TEALS program.
- Assisted students with learning programming fundamentals and assisted during debugging sessions.
- Helped students work through programming exercises by breaking down errors and guiding them through step-by-step problem solutions.